

Hanlong Chen

Curriculum Vitae

Contact Information

Email: sl.chen@mail.utoronto.ca

Website: hanlongchen.com

Education

PhD in Biostatistics, University of Toronto

Sep 2026 - Aug 2030

- Supervisor: Dr. Aya Mitani

MSc in Biostatistics, University of Toronto

Sep 2024 - Jun 2026

- Supervisor: Dr. Aya Mitani
- Thesis: Dynamic Prediction of Tooth Loss Using Landmarking Models with Clustered Survival Data.
- Emphasis in Artificial Intelligence and Data Science.

Honours BSc in Mathematics & Statistics, University of Toronto *Sep 2019 - Jun 2024*

Publications and Preprints

†: Corresponding Author;

*: Equal contribution

Under Review

1. Aya Mitani, **Hanlong Chen**, Yuning Wang and Christopher McCulloch *2026*
Assessing the Reliability of Electronic Dental Records Through Agreement of Probing Pocket Depths Measured Between Two Closely Conducted Periodontal Exams.

Submitted to Journal of Clinical Periodontology

2. **Hanlong Chen**, Xinyang Feng and Aya Mitani.

Dynamic Prediction of Tooth Loss Using Landmarking Models with Clustered Survival Data.

Passed thesis defense, pending to submit

In Progress

1. **Hanlong Chen**^{*,†}, Ian Zhang^{*}, ..., Malcolm Binns *2026*
Continuous Cognitive Heterogeneity in Dementia: A Severity Factor and an Age-Related Amnesic Axis Without Reproducible Subtypes

2. Ian Zhang^{*,†}, **Hanlong Chen^{*}**, ..., Malcolm Binns 2026
 Redundancy-Aware Model Assessment for Component and Cluster Selection in Clinical Subtype Discovery
3. **Hanlong Chen**, Bruce Liu, Aya Mitani, Karim Keshavjee, Felipe Cepeda, Aziz Guer-gachi, and Camellia Zakaria. 2026
 A Sequential Mixed-Effects Imputation Framework for Visit-Level Informative Missingness and Imputation-Aware Evaluation of Metabolic Subtyping Methods in Longitudinal EHR Data.

Research Experience

Canadian Primary Care Sentinel Surveillance Network Toronto, ON
Research Assistant Apr 2025 – Present
 Supervisor: Dr. Camellia Zakaria & Dr. Aya Mitani

Baycrest, Rotman Research Institute Toronto, ON
Graduate Practicum Student Oct 2024 – Present
 Supervisor: Dr. Malcolm Binns

University of Toronto, Department of Statistical Sciences Toronto, ON
Undergraduate Research Student Sep 2023 – May 2024
 Supervisor: Dr. Murari Singh

Internship

Guangzhou Center for Disease Control and Prevention Guangzhou, China
Analyst Intern May – Aug 2021 & May – Aug 2023
 Supervisor: Yilie Ma

Guangzhou Baiyun International Airport Guangzhou, China
Data Analyst Intern Jan 2021 – Apr 2021

Teaching Assistant

University of Toronto Toronto, ON
St. George Campus

- AI4PH - Handling Missing Data in Health Research Winter 2026
- STA130: Introduction to Statistical Reasoning and Data Science Fall 2025
- COG260: Data, Computation, and the Mind Winter 2026

Scarborough Campus

- STAB22: Statistics I Summer 2026
- MATB44: Differential Equations I Winter 2026
- STAA57: Introduction to Data Science Winter 2026
- STAB27: Statistics II Winter 2026
- STAC33: Introduction to Applied Statistics Winter 2026
- STAD91: Bayesian Statistical Analysis Winter 2026

Presentations

Conference Talks

2026 Joint Statistical Meetings *2026*

Hierarchical Gaussian Process Landmarking and Cluster-Aware Metrics for Tooth-Level Survival.

2026 SSC Annual Meeting in Hamilton *2026*

Landmark Models for Clustered Data to Account for Within-cluster Correlation of Unit.

2025 SSC Annual Meeting in Saskatoon *2025*

Cognitive Profiles of People Living with Dementia: a PCA-Based Clustering Analysis.

Posters

The Thirteenth Annual Canadian Statistics Student Conference *2025*

Cognitive Profiles of People Living with Dementia: a PCA-Based Clustering Analysis.

Grants

Graduate Student Research Fellowship *Jun - Sep 2026*

Department of Political Science at the University of Toronto \$3,000

SGS Conference / Research Travel Grant *May 2026*

School of Graduate Study at the University of Toronto \$500

Graduate Student Research Fellowship *Feb 2026*

Department of Psychology at the University of Toronto \$1,000

Jack and Rita Catherall Fund – Research Grant *May 2025*

Research Training Centre at Baycrest \$500

Services

Biostatistics Union of Graduate Students Executive Team *2025 - 2026*

Seminar Committee

Other

Languages Spoken: English (bilingual), Mandarin (native), Cantonese (native)

Memberships: Statistical Society of Canada